Student Name: _____

Student Email: _____



Quiz 0: Starting Points (CS 7150)
Fall 2026

What this is

- **This is not a test, and correctness is not graded.** You get full credit for handing it in with an attempt at every part. Nothing here affects your grade beyond that.
- Its only purpose is to tell me what this particular class already knows, so that I teach the right things. If a whole section comes back blank, I will cover that material. If you guess and I cannot tell what you do not know, I will teach the wrong things.
- If you do not know something, **write "don't know"** and move on. That answer is genuinely more useful to me than a guess.
- You have 15 minutes. Closed book, no calculator, no phone, no laptop, no AI. Every answer here is designed to be worked out by hand.

1. Multiply by hand.

   (a) $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} =$ ☐

   Solution: $[6, 4]^\top$.

2. Least squares. You want to fit $y \approx Ax$ where A is 100×3, y has 100 entries, and x has 3.

   (a) What quantity does the least-squares solution minimize? Write it. ☐

   Solution: $\|Ax - y\|^2$, the sum of squared residuals.

3. Derivatives. Let $u = wx + b$ with x and b fixed, and let $z = g(u)$ for some differentiable g.

   (a) Write $\dfrac{dz}{dw}$ using the chain rule. ☐

4. Expectation and variance. Let X be 1 with probability p and 0 otherwise.

   (a) $\mathbb{E}[X] =$ ☐ (b) $\mathrm{Var}(X) =$ ☐

5. Conditional probability. A disease affects 1% of a population. A test is positive for 99% of people who have it, and for 5% of people who do not. A randomly chosen person tests positive.

(a) What is the probability that they have the disease?

Leave as a fraction if you like. $P(\text{disease}|\text{positive}) =$ [ ]

Solution: $\dfrac{(0.01)(0.99)}{(0.01)(0.99)+(0.99)(0.05)} = \dfrac{0.0099}{0.0594} = \dfrac{1}{6} \approx 17\%$.

6. Capacity. For each, circle whether making the model *more* flexible generally increases or decreases it:

(a) Bias. increases decreases

Solution: Decreases.

(b) Variance. increases decreases

Solution: Increases.

(c) Training error. increases decreases

Solution: Decreases.

7. Rate your comfort, 1 (never seen it) to 5 (very comfortable):

(a) Linear algebra: matrices, eigenvectors, SVD 1 2 3 4 5

(b) Probability: Bayes rule, expectation, maximum likelihood 1 2 3 4 5

(c) Multivariable calculus: gradients, the chain rule 1 2 3 4 5

(d) Python and numpy 1 2 3 4 5

(e) PyTorch (or TensorFlow / JAX) 1 2 3 4 5

(f) Reading a research paper end to end 1 2 3 4 5

Solution: Unscored. Compare against the measured sections: the gap between claimed and demonstrated comfort is itself informative, and in my experience PyTorch is over-reported and probability is under-reported.

8. Yes or no.

(a) Have you trained a neural network yourself (any framework, any size)? yes no

(b) Have you run code on a GPU? yes no

(c) Do you have a laptop that can run Python for this course? yes no

(d) Have you used Google Colab? yes no

Solution: Unscored, but (c) and (d) are logistics you want to know on day one: the homeworks assume Colab or a local Python, and HW5's second notebook needs a GPU runtime.

9. Anything you would like me to know about your background, or anything you are hoping to get out of this course?

Solution: Unscored.